How Do We Formalize a Nonlinear Statistical Dependency Test ?

Let $\mathcal{F}$ and $\mathcal{G}$ be a characteristic RKHS

Regularization Constraint

$$\sup_{g \in \mathcal{G}} \sup_{f \in \mathcal{F}} \mathbf{Cov}(f(X), g(S)) = \langle g, \mathbb{C}\mathbf{ov} f \rangle_{\mathcal{G}} \qquad s.t \qquad \|f\|_{\mathcal{F}} = \|g\|_{\mathcal{G}} = 1$$

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$\|\mathbb{C}\mathrm{ov}\|^2_{HS} = 0 \iff$ Statistical Independence